A. (a) Consider

(b) We can define $\alpha : \mathbb{R} \to \mathbb{R}^3$ as

$$\alpha(t) = (f(t), f(-t), 42)$$

where

$$f(t) = \begin{cases} 0 & \text{if } t \leq 0 \\ e^{-t^{-1}} & \text{if } t > 0 \end{cases}.$$

B. (a) The issue with Gary's proof is that it is not guaranteed that γ is on S. γ must be on S on a small interval $(-\epsilon, \epsilon) \subseteq \mathbb{R}$, but γ is only guaranteed to be on S when $t = 0$.

(b) *Proof.* Let $X \in T_pS$. Then, we have an $\alpha : (a, b) \to \mathbb{R}^n$ in S such that $\alpha(0) = p$ and $\dot{\alpha}(0) = X$. Consider $\tilde{\alpha} : (\frac{a}{c}, \frac{b}{c}) \to \mathbb{R}^n$ given by $\tilde{\alpha}(t) = \alpha(ct)$, $c \neq 0$. We see that $\tilde{\alpha}(0) = p$. We compute that $\dot{\tilde{\alpha}}(t) = c \cdot \dot{\alpha}(ct)$. So, when $t = 0$, $\dot{\tilde{\alpha}}(0) = c \cdot \dot{\alpha}(0) = cX$. We now show that we also have $\mathbf{0}$. We know we have $\tilde{\alpha}(t) = \alpha(-t)$. So, we have $-X \in T_pS$. We are given that T_pS is closed under addition, so $X - X = 0 \in T_pS$. We are done. □

C. *Proof.* When $\mu = (0, 0, 1)$, $k\mu \cdot x = k \cos u$. So, $f(x) = ce^{k \cos u}$. We compute

$$c \iint_{\mathbb{S}^2} e^{k \cos u} \sin u \, \mathrm{d}u \, \mathrm{d}v = c \int_{-\pi}^{\pi} \int_{0}^{\pi} e^{k \cos u} \sin u \, \mathrm{d}u \, \mathrm{d}v = \frac{2c\pi(e^k - \frac{1}{e^k})}{k}.$$

using a u-substitution where $u = \cos u$. In order to make this distribution physically valid, we must solve for c such that

$$\frac{2c\pi(e^k - \frac{1}{e^k})}{k} = 1.$$

Note that $e^k - \frac{1}{e^k} = e^k - e^{-k}$ which is equivalent to $\sinh k$. So, we find that $c = \frac{k}{4\pi \sinh k}$. □